

## 第2讲. 多元函数及极限.

作业: P23. 1 (2). (4). (5). (6). (7). (8). (9). (11).

2 (2). (4). (5). 3 (2). (3). (5). 5 (1). (2). (5). 6.

一. 多元函数.

$$y = f(x) \quad x \in I \subset \mathbb{R}.$$

$$\downarrow$$

$$y = f(x_1, x_2, \dots, x_n).$$

$$(x_1, x_2, \dots, x_n) \in \mathbb{R}^n \xrightarrow{f} \mathbb{R}.$$

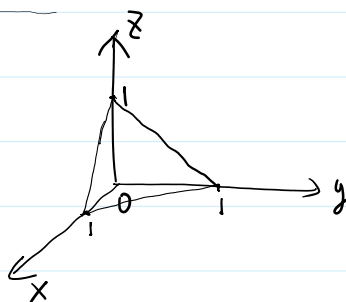
$$z = f(x, y) \quad (x, y) \in D \subseteq \mathbb{R}^2.$$

$$\left\{ (x, y, z) \mid z = f(x, y), (x, y) \in D \right\}$$

例1.  $z = ax + by + c.$   $(x, y) \in \mathbb{R}^2.$ 

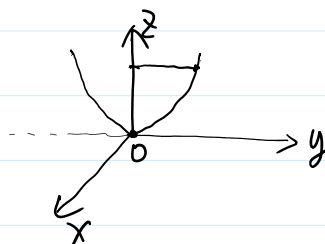
平面.

$$x + y + z = 1$$

例2.  $z = x^2 + y^2.$   $\mathbb{R}^2.$ 

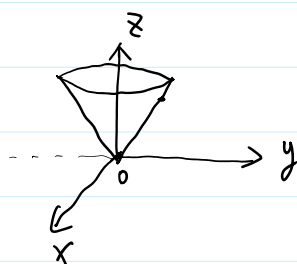
旋转抛物面.

$$z = x^2 + y^2$$

例3.  $z = \sqrt{x^2 + y^2}.$   $\mathbb{R}^2.$ 

圆锥面.

$$z = |y|$$



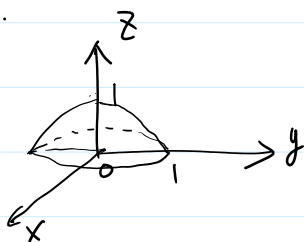
例4.

$$D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$$

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例4.  $z = \sqrt{1-x^2-y^2}$   $D = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \leq 1\}$

$$x^2+y^2+z^2=1.$$



例5.  $z = \frac{x^2}{a^2} - \frac{y^2}{b^2}$   $\mathbb{R}^2$ .

双曲抛物面. (马鞍面).

例6.  $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$

例7.  $z = \sqrt{\frac{x^2}{a^2} + \frac{y^2}{b^2}}$   $\mathbb{R}^2$ .

= 二元函数在一点极限. (=重极限)

定义1. 设  $f(x,y)$  定义在  $D \subset \mathbb{R}^2$  上.  $P_0(x_0, y_0)$  是  $D$  的聚点.  $a \in \mathbb{R}$  给定.

若  $\forall \varepsilon > 0, \exists \delta > 0$ , s.t.  $\forall P(x,y) \in D$  且  $0 < \|PP_0\| < \delta$ .

有  $|f(x,y) - a| < \varepsilon$ .

称  $f(x,y)$  在  $(x,y) \rightarrow (x_0, y_0)$  存在极限  $a$ . 记  $\lim_{(x,y) \rightarrow (x_0, y_0)} f(x,y) = a$ .

$(x,y) \in D$

①.  $P_0(x_0, y_0)$  是  $D$  内点.  $\lim_{(x,y) \rightarrow (x_0, y_0)} f(x,y) = a$ .

例1. 设  $f(x,y) = \frac{x^3-y^3}{x^2+y^2}$   $((x,y) \neq (0,0))$ . 证明:  $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$ .

证明:  $\forall \varepsilon > 0$ , 取  $\delta = \frac{\varepsilon}{\sqrt{2}}$ .  $\forall (x,y), 0 < \sqrt{x^2+y^2} < \delta$

$$|f(x,y) - 0| = \left| \frac{x^3-y^3}{x^2+y^2} \right| \leq \left( \left| \frac{x^2}{x^2+y^2} \right| \right) |x| + \left( \left| \frac{y^2}{x^2+y^2} \right| \right) |y|$$

$$\leq |x| + |y| \leq \sqrt{2} \cdot \sqrt{x^2+y^2} < \varepsilon$$

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$$\therefore \lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0.$$

解法 = . 令  $\begin{cases} x = \rho \cos \theta \\ y = \rho \sin \theta \end{cases} \quad \theta \in [0, 2\pi].$

$$(x,y) \rightarrow (0,0) \Rightarrow \rho \rightarrow 0$$

$$0 \leq |f(x,y)| = \left| \frac{\rho^3 (\sin^2 \theta - \cos^2 \theta)}{\rho^2} \right| \leq 2\rho \rightarrow 0 \quad (\rho \rightarrow 0)$$

$$\therefore \lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0.$$

练习1. 设  $f(x,y) = \frac{(y-x)x}{\sqrt{x^2+y^2}} \quad ((x,y) \neq (0,0))$

证明:  $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0.$

例2. 设  $f(x,y) = \frac{e^{x^2+y^2} - 1}{x^2+y^2} \quad ((x,y) \neq (0,0))$

讨论  $\lim_{(x,y) \rightarrow (0,0)} f(x,y) ?$

解: 令  $\begin{cases} x = \rho \cos \theta \\ y = \rho \sin \theta \end{cases} \quad \theta \in [0, 2\pi].$

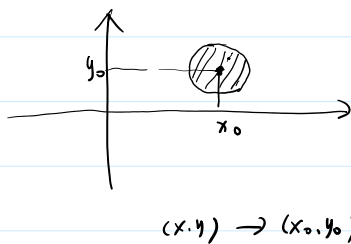
$$0 \leq |f(x,y)| = \left| \frac{e^{\rho^2(\sin^2 \theta + \cos^2 \theta)} - 1}{\rho^2} \right| = |O(\rho)| \rightarrow 0$$

$$e^x = 1 + \underbrace{(x + o(x))}_{(x \rightarrow 0)} = x(1 + \frac{1}{2}x + \frac{1}{6}x^2 + \dots) \quad (\rho \rightarrow 0)$$

$$e^x - 1 = O(x).$$

$$|e^x - 1| \leq M|x|$$

$$\therefore \lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0.$$



例3. 设  $f(x,y) = \frac{xy}{\sqrt{x+y+1} - 1}$   $(x,y) \neq (0,0)$

讨论  $\lim_{(x,y) \rightarrow (0,0)} f(x,y) ?$

解:

$$f(x,y) = \frac{xy}{x+y} \left( \underbrace{\sqrt{x+y+1} + 1}_2 \right)$$

$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x+y}$  不存在.

$(x,y) \rightarrow (0,0) \Leftrightarrow x^2 + y^2 \rightarrow 0$

$$\begin{cases} x=0, y \rightarrow 0 \\ x \rightarrow 0, y=0 \\ x \rightarrow 0, y \rightarrow 0 \end{cases}$$

$y = kx$ .  $k \neq -1$ .

$$\frac{kx}{(1+k)} \rightarrow 0 \quad (x,y) \rightarrow (0,0)$$

X

$x+y = kx^2$   $k \neq 0$ .

$$x(kx^2 - x) = x^2(kx - 1)$$

$$\frac{xy}{x+y} = \frac{kx-1}{k} \rightarrow -\frac{1}{k} \quad ((x,y) \rightarrow (0,0))$$

$f(x,y) = \frac{xy}{x+y}$   $(0,0)$

$$\forall y \neq 0,$$

$$f(x, y) = \frac{xy}{x+y} \rightarrow 0, \quad (x \rightarrow 0)$$

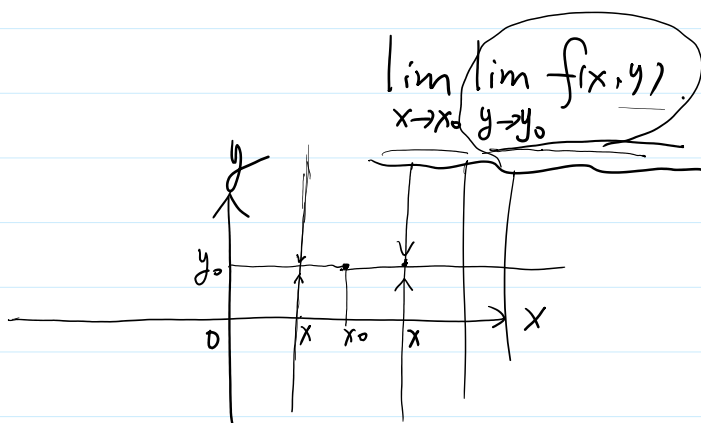
$$\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} f(x, y) = 0. \quad \text{累次.}$$

定义 2.  $(x_0, y_0)$ .

$$\forall \text{ fix. } y \neq y_0. \quad \text{若 } \lim_{x \rightarrow x_0} f(x, y) = \varphi(y) \text{ 存在.}$$

$$\text{且 } \lim_{y \rightarrow y_0} \varphi(y) = A \text{ 存在.} \quad \text{则 } \lim_{y \rightarrow y_0} \lim_{x \rightarrow x_0} f(x, y) \text{ 称为 } f(x, y) \text{ 在 } (x_0, y_0) \text{ 处对 } x \text{ 后对 } y$$

累次极限.



$$\text{例 4. } f(x, y) = x \sin \frac{1}{y} \quad (y \neq 0). \quad (0, 0)$$

$$0 \leq |f(x, y)| \leq |x| \rightarrow 0 \quad ((x, y) \rightarrow (0, 0))$$

$$\therefore \lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0.$$

$$x \neq 0, \quad \lim_{x \rightarrow 0} \lim_{y \rightarrow 0} x \sin \frac{1}{y} \text{ 不存在.}$$

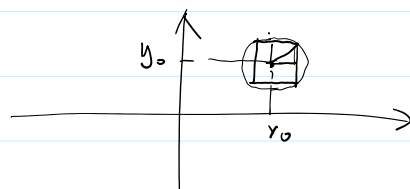
$$\forall y \neq 0. \quad \lim_{y \rightarrow 0} \lim_{x \rightarrow 0} x \sin \frac{1}{y} = 0.$$

定理 1. 设  $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = A$  存在 且  $\forall x \neq x_0, \quad \lim_{y \rightarrow y_0} f(x, y)$  存在.

$$b1 \quad 1. \quad 1. \quad \lim_{n \rightarrow \infty} - A$$

$$k) \lim_{x \rightarrow x_0} \lim_{y \rightarrow y_0} f(x, y) = A.$$

$$\text{证: } \therefore \lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = A.$$



$$\therefore \forall \varepsilon > 0. \quad \exists \delta > 0. \quad \text{s.t.} \quad \forall (x, y).$$

$$0 < |x - x_0| < \frac{\delta}{\sqrt{2}} \quad 0 < |y - y_0| < \frac{\delta}{\sqrt{2}}$$

$$\therefore 0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta.$$

$$|f(x, y) - A| < \varepsilon.$$

$$\forall x. \quad 0 < |x - x_0| < \frac{\delta}{\sqrt{2}}.$$

$$\lim_{y \rightarrow y_0} |f(x, y) - A| \leq \varepsilon.$$

||

$$\left| \lim_{y \rightarrow y_0} f(x, y) - A \right|$$

||  
(f(x))

$$\therefore \lim_{x \rightarrow x_0} \lim_{y \rightarrow y_0} f(x, y) = A.$$

定理 2. 设  $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = \infty$ . 且  $\forall x \neq x_0$ .  $\lim_{y \rightarrow y_0} f(x, y)$  存在.

$$k) \lim_{x \rightarrow x_0} \lim_{y \rightarrow y_0} f(x, y) = \infty.$$

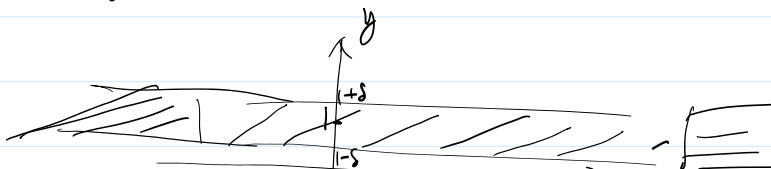
$$\begin{cases} x \rightarrow a \\ y \rightarrow \infty \end{cases}$$

$$\begin{cases} x \rightarrow \infty \\ y \rightarrow b \end{cases}$$

$$\begin{cases} x \rightarrow \infty \\ y \rightarrow \infty \end{cases}$$

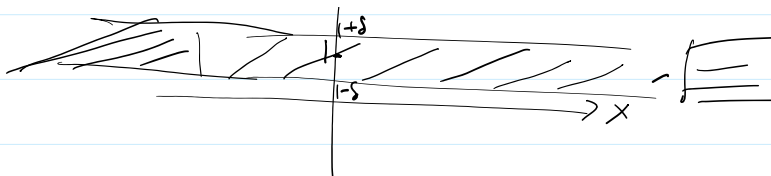
例 5.

$$\text{求 } \lim_{x \rightarrow \infty} \left(1 + \frac{1}{xy}\right)^{\frac{x^2}{x+y}}$$



“1”  
求

$$\lim_{\substack{x \rightarrow \infty \\ y \rightarrow 1}} \left(1 + \frac{1}{xy}\right)^{\frac{x^2}{x+y}}$$



$$\lim_{\substack{x \rightarrow \infty \\ y \rightarrow 1}} \left(1 + \frac{1}{xy}\right)^{xy} \cdot \frac{x^2}{(x+y)xy} = \frac{1}{y + \frac{y^2}{x}}$$

$$\lim_{\substack{x \rightarrow \infty \\ y \rightarrow 1}} \left(1 + \frac{1}{xy}\right)^{xy} \cdot \lim_{\substack{x \rightarrow \infty \\ y \rightarrow 1}} \frac{1}{y + \frac{y^2}{x}}$$

||  
e

1/11.  $\lim_{x \rightarrow x_0} u(x) = a > 0$

$\lim_{x \rightarrow x_0} v(x) = b$

2.  $\lim_{x \rightarrow x_0} u(x)^{v(x)} = a^b$

例 6. 求  $\lim_{\substack{x \rightarrow \infty \\ y \rightarrow \infty}} \frac{(x^2 + y^2) \sin(x^4 + y^4)}{x^4 + y^4} = 0$

解:

$$0 \leq \left| \frac{(x^2 + y^2) \sin(x^4 + y^4)}{x^4 + y^4} \right| \leq \left| \frac{x^2 + y^2}{x^4 + y^4} \right|$$

$$\begin{cases} x = \rho \cos \theta \\ y = \rho \sin \theta \end{cases}$$

$x \rightarrow \infty$   
 $y \rightarrow \infty \Rightarrow \rho \rightarrow \infty$

$$\frac{1}{\rho^2 (\sin^4 \theta + \cos^4 \theta)}$$

$$\frac{1}{\rho^2 (1 - \frac{1}{2} \sin^2 2\theta)} \leq \frac{2}{\rho^2} \rightarrow 0$$

$$\lim_{(x,y) \rightarrow (1,0)} \frac{\sin xy}{y-x} \cdot x = 1$$